

where F is any anti-derivative (or indefinite integral) of continuous function f on a closed interval $[a, b]$. This part of the theorem is one that we use is finding area. It allows us to compute the definite integral of a function by using one of infinitely many anti-derivatives, or indefinite integrals. Here is a simple example of the fundamental theorem of calculus:

$$\int_0^1 \cos(t) dt = \sin(t) \Big|_0^1 = \sin(1) - \sin(0) = \sin(1)$$

It is important to note that if $F(x)$ is an anti-derivative of $f(x)$ and the function is defined on some interval, then every other anti-derivative $G(x)$ of $f(x)$ differs from $F(x)$ by a constant:

$$G(x) = F(x) + C$$

If we recall the general power rule for integration, taking the integral of a function will give us a constant. So, there are infinitely many different anti-derivatives for any given function.

12.9.1 Evaluation of Definite Integrals

Let us now consider a few examples using the fundamental theorem of calculus.

Example 12.25 : Evaluate: $\frac{d}{dx} \int_a^x 4t^3 dt$

$$\text{Ans: } \frac{d}{dx} \int_a^x 4t^3 dt = \frac{d}{dx} (t^4)$$

Use the second part of the theorem and solve for the interval $[a, x]$

$$\frac{d}{dx} (x^4 - a^4) = 4x^3$$

Taking the derivative with respect to x will leave out the constant giving $f(x) = 4x^3$.

Example 12.26 : Evaluate: $\frac{d}{dx} \int_2^{\sin(x)} e^{t^2} dt$

Ans: We need to find out the derivative of the integrand from the interval $[2, \sin(x)]$. In other words, From the first part of the fundamental theorem of calculus, we get the following:

$$\text{Let } G(x) = \int_2^x e^{t^2} dt \text{ Then } G'(x) = e^{x^2}$$

Example 12.27 : Evaluate: $\int_2^3 x^2 dx$

$$\text{Ans: } \int_2^3 x^2 dx = x^3 \Big|_2^3 = 3^3 - 2^3 = 19$$

Example 12.28: Evaluate $\int_0^2 (t^2 - 3t + 10) dt$

$$\text{Ans: } \int (t^2 - 3t + 10) dt = \frac{1}{3}t^3 - \frac{3}{2}t^2 + 10t + c$$

$$\text{So, } \int_0^2 (t^2 - 3t + 10) dt = \left(\frac{8}{3} - \frac{12}{2} + 20 \right) - 0 = \frac{50}{3}$$

Check Your Progress 4

- 1) What do you understand by fundamental theorem of calculus?

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- 2) Differentiate between integration and differentiation.

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- 3) Distinguish between definite and infinite integrals.

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12.10 INTEGRATION WITH CHAIN RULE

We now discuss the area between the graph of $y = f(x)$ and the x - axis by the definite integral (Fig. 12.7). You will see it gives a positive result for a graph above the x - axis, and a negative result for a graph below the x - axis.

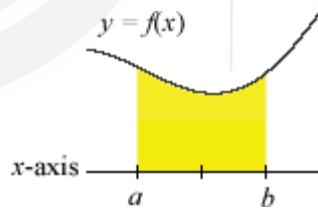


Fig. 12.7: Area between the graph of $y = f(x)$ and x -axis

If the graph of $y = f(x)$ is partly above and partly below the x - axis, the formula given below generates the net area (Fig 12.8). We get the area above the axis minus the area below the axis.

Example 12.29: Find the area between $y = 7 - x^2$ and the x -axis between the values $x = -1$ and $x = 2$ (Fig. 12.8).

$$\begin{aligned} \text{Ans: Area} &= \int_{-1}^2 (7 - x^2) dx = \left(7x - \frac{1}{3}x^3 \right) \Big|_{-1}^2 \\ &= \left[7 \cdot 2 - \frac{1}{3}(8) \right] - \left[7(-1) - \frac{1}{3}(-1) \right] = 18 \end{aligned}$$

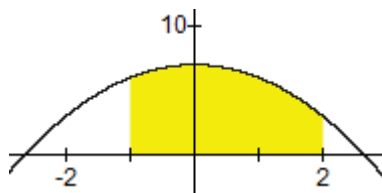


Fig. 12.8: Area between $y = 7 - x^2$ and ordinates corresponding to different values of x

Example 12.30: Find the net area between $y = \sin x$ and the x -axis between the values $x = 0$ and $x = 2\pi$.

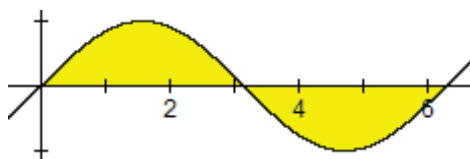


Fig. 12.9: Area of curve which is above and below x -axis

Ans: Net Area = $\int_0^{2\pi} \sin x \, dx$ **NetArea** = $\int_0^{2\pi} \sin x \, dx$
 $= (-\cos x)|_0^{2\pi} = (-1) - (-1) = 0.$

However, the actual area can be calculated by taking the sum of the positive values of the following two integrals $\int_0^{\pi}$ and $\int_{\pi}^{2\pi}$

12.10.1 Integration with Chain Rule and Substitution

Next, we briefly discuss how to change limits in definite integrals in substitution process. When evaluating a definite integral using u -substitution, how does one deal with the limits of integration? There are two possibilities. As before, we have with an integral of the form $\int_a^b f(x)f'(x)dx$. We will again let $u = f(x)$, giving

us $du = f'(x) dx$. When we make the substitution in the integral, the limits of integration also change, giving us $\int_{u_1}^{u_2} u du$. Normal integration, as before, gives

us $\int_{u_1}^{u_2} u du = \left[\left(\frac{u^2}{2} \right) \right]_{u_1}^{u_2}$. But what are these new limits of integration u_1 and u_2 ? We

can regard them simply as place holders. When we reverse our replacement, substituting $\sin x$ for u , the limits of integration also revert to the original a and b .

This gives us $\int_{u_1}^{u_2} u du = \left[\left(\frac{u^2}{2} \right) \right]_{u_1}^{u_2} = \left[\left(\frac{\sin^2 x}{2} \right) \right]_a^b$, which can be evaluated in the

standard fashion. Alternatively, after finding $\int_{u_1}^{u_2} u du = \left[\left(\frac{u^2}{2} \right) \right]_{u_1}^{u_2}$, we could use the

fact that since $u = f(x)$, $u_2 = f(b)$ and $u_1 = f(a)$, to evaluate directly without

reversing the u-substitution. Using the method, we have

$$\int_a^b f(x)f'(x)dx = \int_{u_1}^{u_2} u du = \left[\left(\frac{u^2}{2} \right) \right]_{f(a)}^{f(b)} = \frac{1}{2} \left((f(b))^2 - (f(a))^2 \right).$$

Example 12.31 : Evaluate the following using u-substitution:

$$\int_0^2 2x\sqrt{4-x^2} dx$$

Ans. Let $u = (4 - x^2)$, then $du = -2x dx$

$$\int_0^2 2x\sqrt{4-x^2} dx = - \int_{u_1}^{u_2} (\sqrt{u}) du = - \left[\left(\frac{2}{3} (u)^{\frac{3}{2}} \right) \right]_{u_1}^{u_2} = - \left[\left(\frac{2}{3} (4-x^2)^{\frac{3}{2}} \right) \right]_0^2 = -\frac{2}{3} (0-8) = 16/3$$

Example 12.32: Evaluate $\int_0^{\frac{\pi}{2}} \sin x \cos x dx$.

Ans. *Method 1:* Put $u = \sin x$ to get $du = \cos x dx$.

On substitution of u and du the problem reduces to $\int_{u_1}^{u_2} u du$ integrating, reversing the substitution, and evaluating, we get

$$\left(\frac{u}{2} \right) \Big|_{u_1}^{u_2} = \left(\frac{\sin^2}{2} \right) \Big|_0^{\pi/2} = \frac{1}{2} \left[\left(\sin^2 \left(\frac{\pi}{2} \right) \right) \right] - \sin^2(0) = \frac{1}{2} (1-0) = \frac{1}{2}$$

Method 2: As in method 1, we substitute and integrate to find

$$\int_0^{\frac{\pi}{2}} \sin x \cos x dx = \int_{u_1}^{u_2} u du = \left\langle \frac{u^2}{2} \right\rangle \Big|_{u_1}^{u_2}.$$

Since $u = f(x), u_1 = f(0) = \sin(0) = 0$ and $u_2 = f\left(\frac{\pi}{2}\right) = \sin\left(\frac{\pi}{2}\right) = 1$. Thus, we have

$$\left\langle \frac{u^2}{2} \right\rangle \Big|_{u_1}^{u_2} = \frac{1^2}{2} - \frac{0^2}{2} = \frac{1}{2}, \text{ which is the same as the answer we found using method 1.}$$

Example 12.33: Evaluate $\int_0^{\frac{\pi}{2}} \sin^2 x \cos x dx$.

Ans. We make the same substitution as in examples 1 and 2, $u = \sin x$ $du = \cos x dx$. This gives the integral $u^2 du$, which can be evaluated normally exactly as

$$\int_{u_1}^{u_2} u du. \text{ In this case, we find}$$

$$\int_{u_1}^{u_2} u^2 du = \left\langle \frac{u^2}{3} \right\rangle \Big|_{u_1}^{u_2} = \left\langle \frac{\sin^3}{3} \right\rangle \Big|_0^{\frac{\pi}{2}} = \frac{1}{3} \left\langle \sin^3 \left\langle \frac{\pi}{2} \right\rangle - \sin^3(0) \right\rangle = \frac{1}{3}(1-0) = \frac{1}{3}$$

Check Your Progress 5

- 1) What is the relation between integration and differentiation with respect to the use of chain rule?

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- 2) Suppose you are evaluating, integral $\int_{-1}^2 4x(2x^2 + 3)^2 dx$ by u-substitution. If you take $u = 2x^2 + 3$, what are the new limits?

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- 3) Evaluate the integral, $\int_{-1}^2 4x(2x^2 + 3)^2 dx$ using u-substitution.

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12.11 IMPROPER INTEGRALS

We now discuss briefly one special type of definite integral, the **improper integral**. When one of the limits of integration is $+\infty$ or $-\infty$ a definite integral is called an improper integral. Such integrals are evaluated using the concept of limits according to the following rules:

$$\begin{aligned} \text{i)} \quad & \int_a^\infty f(x) dx = \lim_{b \rightarrow \infty} \int_a^b f(x) dx \\ \text{ii)} \quad & \int_{-\infty}^b f(x) dx = \lim_{a \rightarrow -\infty} \int_a^b f(x) dx \end{aligned}$$

Example 12.34: Evaluate $\int_1^\infty \frac{dx}{x^2}$

Ans. Since, $\int_1^b \frac{dx}{x^2} = -\frac{1}{b} + 1$, the desired integral is

$$\lim_{b \rightarrow \infty} \int_1^b \frac{dx}{x^2} = \lim_{b \rightarrow \infty} \left(-\frac{1}{b} \right) = 1.$$

This integral can be interpreted as the area of the unbounded region between $f(x) = \frac{1}{x^2}$, $y = 0$ (the x -axis), and $x = 1$.

12.11.1 Convergent and Divergent Integrals

Definition: Let f be a function which is continuous on the closed interval $[b, \infty[$. We define

$$\int_{-\infty}^b f(x) dx = \lim_{N \rightarrow -\infty} \int_N^b f(x) dx$$

If this limit exists and is finite then we say that the integral $\int_{-\infty}^b f(x) dx$ is convergent; otherwise, we say that the integral is divergent.

Example 12.35: Determine if the integral $\int_1^{\infty} \frac{1}{x} dx$ is convergent or divergent and if it is convergent find its value.

Ans: So, the first thing we do is convert the integral to a limit.

$$\int_1^{\infty} \frac{1}{x} dx = \lim_{a \rightarrow \infty} \int_1^a \frac{1}{x} dx$$

Now, do the integral and the limit.

$$\int_1^{\infty} \frac{1}{x} dx = \lim_{a \rightarrow \infty} \ln(a) - \ln(1) = \infty$$

So, the limit is infinite and the integral is divergent.

Example 12.36: Determine if the integral $\int_{-\infty}^0 \frac{1}{\sqrt{3-x}} dx$ is convergent or divergent.

If it is convergent find its value.

Ans: There really is not much to do with these problems once you know how to do them. We will convert the integral to a limit/integral pair, evaluate the integral and then the limit.

$$\begin{aligned} \int_{-\infty}^0 \frac{1}{\sqrt{3-x}} dx &= \lim_{b \rightarrow -\infty} \int_{-\infty}^0 \frac{1}{\sqrt{3-x}} dx = \lim_{b \rightarrow -\infty} -2\sqrt{3-x} \Big|_b^0 = \\ &= \lim_{b \rightarrow -\infty} (-2\sqrt{3} + 2\sqrt{3-b}) = -2\sqrt{3} + \infty \end{aligned}$$

So, the limit is infinite and so this integral is divergent.

Definition: Let f be a function which is continuous for all real numbers. If, for some real number c , both of $\int_{-\infty}^c f(x) dx$ and $\int_c^{\infty} f(x) dx$ are convergent, then we define

$$\int_{-\infty}^{\infty} f(x) dx = \int_{-\infty}^c f(x) dx + \int_c^{\infty} f(x) dx \text{ and we say that the integral } \int_{-\infty}^{\infty} f(x) dx \text{ is}$$

convergent; otherwise, we say that the integral is divergent.

Example 12.37 : Determine if the integral $\int_{-\infty}^{\infty} xe^{-x^2} dx$ is convergent or divergent. If it is convergent find its value.

Ans: Since the problem has infinities in both limits we need to split up the integral into two separate integrals. Let us choose $a = 0$ which is a convenient point for the evaluation process. The integral then is,

$$\int_{-\infty}^{\infty} xe^{-x^2} dx = \int_{-\infty}^0 xe^{-x^2} dx + \int_0^{\infty} xe^{-x^2} dx$$

We can now look at each of the individual limits.

$$\begin{aligned} \int_{-\infty}^0 xe^{-x^2} dx &= \lim_{b \rightarrow -\infty} \int_b^0 xe^{-x^2} dx = \lim_{b \rightarrow -\infty} \left[-\frac{1}{2} e^{-x^2} \right]_b^0 \\ &= \lim_{b \rightarrow -\infty} \left(-\frac{1}{2} + \frac{1}{2} e^{-b^2} \right) = \frac{1}{2} \end{aligned}$$

So, the first integral is convergent. Let us examine the second integral.

$$\begin{aligned} \int_0^{\infty} xe^{-x^2} dx &= \lim_{b \rightarrow \infty} \int_0^b xe^{-x^2} dx = \lim_{b \rightarrow \infty} \left(-\frac{1}{2} e^{-x^2} \right) \Big|_0^b \\ &= \lim_{b \rightarrow \infty} \left(\frac{1}{2} e^{-b^2} + \frac{1}{2} \right) = \frac{1}{2} \end{aligned}$$

Thus, the second integral is convergent and we deal with convergent integral. Its value is,

$$\int_{-\infty}^{\infty} xe^{-x^2} dx = \int_{-\infty}^0 xe^{-x^2} dx + \int_0^{\infty} xe^{-x^2} dx = -\frac{1}{2} + \frac{1}{2} = 0$$

Comparison Theorem: Let f and g be functions which are continuous on the closed interval $[a, \infty)$. Suppose that $f(x) > g(x) > 0$ for all x in $[a, \infty)$.

- i) If $\int_a^{\infty} f(x) dx$ is convergent then $\int_a^{\infty} g(x) dx$ is also convergent.
- ii) If $\int_a^{\infty} g(x) dx$ is divergent then $\int_a^{\infty} f(x) dx$ is also divergent.

Check Your Progress 6

1) What is meant by improper integral?

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- 2) Determine if the integral $\int_{-2}^{\infty} \sin x dx$ is convergent or divergent. If it is convergent, find its value.
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- 3) Determine if the integral $\int_0^3 \frac{1}{(\sqrt{3-x})} dx$ is convergent or divergent. If it is convergent, find its value.
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12.12 INTEGRATION WITH PARTIAL FRACTIONS

Suppose we deal with integrals of rational expressions of polynomials. For example, consider an integral like $\int \frac{3x+11}{x^2-x-6} dx$. You may see that substitution

method will not work here. However, if you break the integrand as, $\frac{3x+11}{x^2-x-6} = \frac{4}{x-3} - \frac{1}{x+2}$, then the integration can be performed as follows:

$$\int \frac{3x+11}{x^2-x-6} dx = \int \left(\frac{4}{x-3} - \frac{1}{x+2} \right) dx = 4 \ln|x-3| - \ln|x+2| + c.$$

Such a decomposition of the rational expression is called **partial fraction decomposition**. Let us have a quick review of such fractions. We will also consider some examples and their solutions for better understanding.

Start with a rational expression of the form: $f(x) = \frac{p(x)}{q(x)}$, where both $p(x)$ and $q(x)$ are polynomials. We can use decomposition to solve the expression if the degree of $p(x)$ is smaller than the degree of $q(x)$. Thus, you need to remember that partial fractions can be considered only if following conditions are satisfied: (i) the rational expression has polynomials in numerator and denominator and (ii) the degree of the numerator is strictly less than the degree of the denominator. After these two conditions are satisfied, we need to find the decomposition for a given function.

For example, a partial fractions decomposition for $\frac{6}{x^2-1}$ can be performed by factoring the denominator such that $\frac{6}{x^2-1} = \frac{6}{(x-1)(x+1)}$. Let us assume that there are constants A and B, so that $\frac{6}{(x-1)(x+1)} = \frac{A}{x-1} + \frac{B}{x+1}$. It can be shown that such constants always exist for the rational function $\frac{p(x)}{q(x)}$ if the above

mentioned two conditions are met. So far, we have

$$\frac{6}{x^2-1} = \frac{6}{(x-1)(x+1)} = \frac{A}{x-1} + \frac{B}{x+1}$$

$$= \frac{A}{x-1} \frac{(x+1)}{(x-1)} + \frac{B}{x+1} \frac{(x-1)}{(x-1)} = \frac{A}{(x-1)} \frac{(x+1)}{(x-1)} + \frac{B}{(x+1)} \frac{(x-1)}{(x-1)} = \frac{A(x+1)+B(x-1)}{(x-1)(x+1)}$$

$$\text{or, } \frac{6}{(x-1)(x+1)} = \frac{A(x+1)+B(x-1)}{(x-1)(x+1)}$$

$$\text{or, } 6 = A(x+1) + B(x-1)$$

Now if we let $x = 1$

$$\text{then } 6 = A(1+1) + B(1-1) = 2A + 0$$

$$\text{so that } A = 3$$

If we choose to let $x = -1$

$$\text{then } 6 = A(-1+1) + B(-1-1) = 0 - 2B$$

$$\text{so that } B = -3$$

$$\text{The decomposition of } \frac{6}{(x-1)(x+1)} \text{ is equal to } \frac{3}{x-1} + \frac{-3}{x+1}$$

Here is another important point to consider when applying the method of partial fractions to the rational function $\frac{p(x)}{q(x)}$. If the degree (highest power) of $p(x)$ is

equal to or greater than the degree of $q(x)$, then you must rewrite the given rational function as the sum of a polynomial and a new rational function satisfying condition (ii) above.

For example, to find a partial fractions decomposition for $\frac{2x-3}{x^3+x}$ begin by

$$\text{factoring the denominator such that } \frac{2x-3}{x^3+x} = \frac{2x-3}{x(x^2+1)}.$$

Assuming that there are three constants A , B and C , write

$$\frac{2x-3}{x(x^2+1)} = \frac{A}{x} + \frac{Bx+C}{x^2+1} = \frac{A(x^2+1)}{x(x^2+1)} + \frac{(Bx+C)x}{x(x^2+1)} = \frac{A(x^2+1)+(Bx+C)x}{x(x^2+1)}$$

$$\text{Thus, } 2x-3 = A(x^2+1) + (Bx+C)x$$

Now if we choose $x = 0$, then $2(0) - 3 = A((0)^2 + 1) + B(0) + C(0) = A$ so that $A = -3$. If we choose to let $x = i$, where i is an imaginary number, then

$$2i - 3 = A(i^2 + 1) + (Bi + C)i = A((-1) + 1) + Bi^2 + Ci = Ci - B$$

so that $B = 3$ and $C = 2$. With these values we get a partial fractions decomposition

$$\text{We can now say that a partial fractions decomposition for } \frac{2x-3}{x(x^2+1)} = \frac{-3}{x} + \frac{3x+2}{x^2+1}.$$

Example 12.38: Let us find partial fractions decomposition of a rational function repeated factors in the denominator.

$$\text{Ans: Let } \frac{2x^3+5x-1}{(x+1)^3(x^2+1)^2} = \frac{A}{x+1} + \frac{B}{(x+1)^2} + \frac{C}{(x+1)^3} + \frac{Dx+E}{x^2+1} + \frac{Fx+G}{(x^2+1)^2}$$

So, once we have determined that partial fractions can be carried out, we factor the denominator as completely as possible. Then for each factor in the denominator we can use determine the term(s) in the partial fraction decomposition.

Example 12.39: Find solution of the integral when degree of the numerator is not less than the degree of the denominator.

$$\text{Integrate } \int \frac{x^4 + x^3 + x^2 + 1}{x^2 + x - 2} dx$$

Ans:

$$\int \frac{x^4 + x^3 + x^2 + 1}{x^2 + x - 2} dx = \int \left(x^2 + 3 + \frac{-3x + 7}{x^2 + x - 2} \right) dx = \int \left(x^2 + 3 + \frac{-3x + 7}{(x + 2)(x - 1)} \right) dx$$

$$\int \left(x^2 + 3 + \frac{A}{x + 2} + \frac{B}{x - 1} \right) dx = A(x - 1) + B(x + 2) = -3x + 7;$$

$$\text{let } x = -2 \text{ so that } A(-3) + B(0) = 13 \rightarrow A = -\frac{13}{3};$$

$$\text{let } x = 1 \text{ so that } A(0) + B(3) = 4 \rightarrow B = \frac{4}{3}$$

$$\begin{aligned} \text{Now } \int \left(x^2 + 3 + \frac{-\frac{13}{3}}{x + 2} + \frac{\frac{4}{3}}{x - 1} \right) dx &= \int \left(x^2 + 3 - \frac{13}{3} \frac{1}{x + 2} + \frac{4}{3} \frac{1}{x - 1} \right) dx \\ &= \frac{x^3}{3} + 3x - \frac{13}{3} \ln|x + 2| + \frac{4}{3} \ln|x - 1| + C. \end{aligned}$$

Example 12.40: Find solution of the integral when there is a repeated linear factor.

$$\text{Integrate } \int \frac{x + 7}{x^2(x + 2)} dx$$

$$\text{Ans: } \int \frac{x + 7}{x^2(x + 2)} dx = \int \left(\frac{A}{x} + \frac{B}{x^2} + \frac{C}{x + 2} \right) dx$$

$$\text{or, } A(x + 2) + B(x + 2) + Cx^2 = x + 7;$$

$$\text{let } x = 0 \text{ so that } A(0) + B(2) + C(0) = 7 \rightarrow B = \frac{7}{2};$$

$$\text{let } x = -2 \text{ so that } A(0) + B(2) + C(4) = 5 \rightarrow C = \frac{5}{4};$$

$$\text{let } x = -1 \text{ so that } A(-1) + B(1) + C(1) = -A + \frac{7}{2} + \frac{5}{4} = 6 \rightarrow A = -\frac{5}{4}$$

$$\text{Now } \int \left(\frac{-\frac{5}{4}}{x} + \frac{\frac{7}{2}}{x^2} + \frac{\frac{5}{4}}{x + 2} \right) dx = \int \left(-\left(\frac{5}{4}\right) \frac{1}{x} + \left(\frac{7}{2}\right) \frac{1}{x^2} + \left(\frac{5}{4}\right) \frac{1}{x + 2} \right) dx =$$